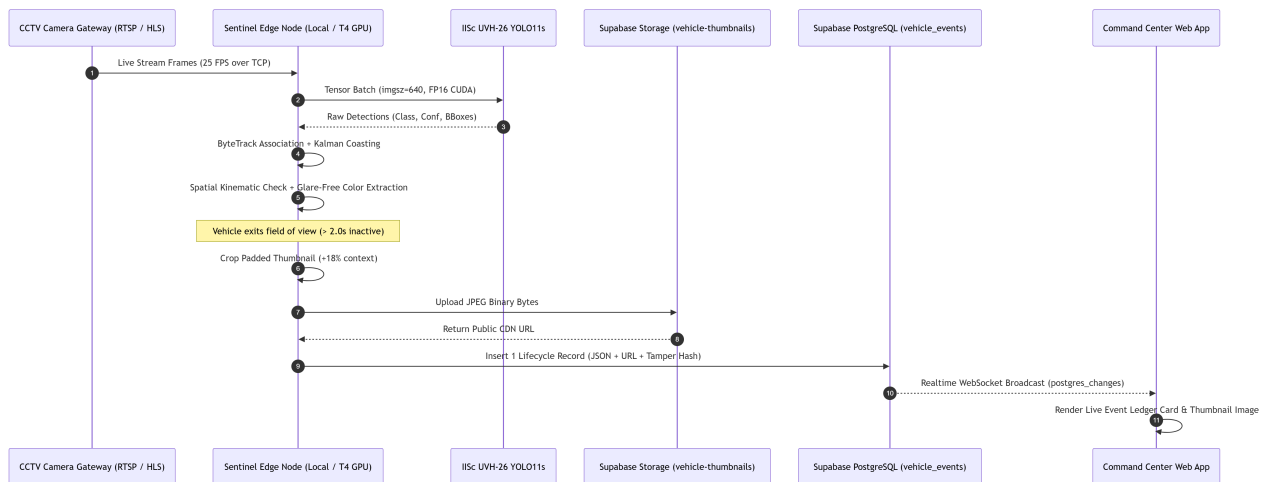


Sentinel (BIG EYE) — Technical Architecture & System Manual

File 2: 02_TECHNICAL_ARCHITECTURE_AND_SYSTEM_MANUAL.md

Technical Architecture & System Manual

1. End-to-End System Topology



2. Edge AI Pipeline Specifications

A. RTSP Stream Synchronization & Watchdog

- **Protocol:** Mandatory RTSP over TCP (`rtsp_transport;tcp`).
- **Connection Watchdog:** Socket timeout capped at 3,000,000 μ s (3.0s). If packet drops exceed 30 consecutive frames, the stream initiates an immediate auto-reconnect or falls back to HLS (`https://cctv.corp8.cloud/<id>/index.m3u8`).
- **Timing Ground Truth:** Driven strictly by Stream Presentation Timestamps (PTS via `CAP_PROP_POS_MSEC`), never host wall-clock time.

B. Tracking & Spatial Kinematic Gating

- **Tracker Core:** Multi-object tracking via ByteTrack with custom-tuned parameters:
 - `track_high_thresh`: 0.38 (Confident primary association)
 - `track_low_thresh`: 0.12 (Recovers dimmed vehicles during 1–3 frame drops)
 - `new_track_thresh`: 0.48 (Prevents entering vehicles from stealing old track IDs)

- `track_buffer`: 60 (Retains Kalman coasting prediction for 2.4s)
- `match_thresh`: 0.78 (Strict spatial overlap)
- **Kinematic Teleportation Filter**: Rejects coordinate displacement jumps > 130 pixels in a single frame.

3. Database Schema & Storage Contract

PostgreSQL Schema (`public.vehicle_events`)

```
CREATE TABLE public.vehicle_events (
  id BIGSERIAL PRIMARY KEY,
  camera_id TEXT NOT NULL,
  track_id INT NOT NULL,
  vehicle_class TEXT NOT NULL,
  dominant_color TEXT,
  motion_state TEXT,
  traffic_flow TEXT,
  net_displacement_px FLOAT DEFAULT 0.0,
  entry_sec FLOAT DEFAULT 0.0,
  exit_sec FLOAT DEFAULT 0.0,
  dwell_sec FLOAT DEFAULT 0.0,
  thumbnail_url TEXT,
  tamper_hash TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Compound indexes for sub-20ms statewide search queries
CREATE INDEX idx_veh_cam ON public.vehicle_events(camera_id);
CREATE INDEX idx_veh_class ON public.vehicle_events(vehicle_class);
CREATE INDEX idx_veh_color ON public.vehicle_events(dominant_color);
CREATE INDEX idx_veh_flow ON public.vehicle_events(traffic_flow);
CREATE INDEX idx_veh_created ON public.vehicle_events(created_at DESC);
```

Storage Bucket Contract

- **Bucket Name**: `vehicle-thumbnails`
- **Visibility**: Public
- **File Key Format**: `<camera_id>_id<track_id>_<vehicle_class>.jpg`
- **Direct URL Endpoint**: `https://<project-ref>.supabase.co/storage/v1/object/public/vehicle-thumbnails/<filename>.jpg`

4. Frontend Command Center Integration

A. Supabase Client Initialization

```
import { createClient } from '@supabase/supabase-js';

const SUPABASE_URL = "https://ulmbrptrssesgptmaewq.supabase.co";
const SUPABASE_ANON_KEY = "eyJhbGciOiI..."; // Project Anon Key

export const supabase = createClient(SUPABASE_URL, SUPABASE_ANON_KEY);
```

B. Live Realtime Stream Listener

```
// Automatically appends new vehicle lifecycle cards without page reload
export function subscribeToTrafficEvents(onNewEvent) {
  return supabase
    .channel('realtime_traffic')
    .on(
      'postgres_changes',
      { event: 'INSERT', schema: 'public', table: 'vehicle_events' },
      (payload) => {
        onNewEvent(payload.new);
      }
    )
    .subscribe();
}
```

C. Public Thumbnail Image Resolver

```
export function getVehicleThumbnailUrl(event) {
  if (event.thumbnail_url && event.thumbnail_url.startsWith('http')) {
    return event.thumbnail_url;
  }
  // Deterministic fallback URL
  return `${SUPABASE_URL}/storage/v1/object/public/vehicle-
  thumbnails/${event.camera_id}_id${event.track_id}_${event.vehicle_class}.jpg
  `;
}
```

5. Local Execution & Runbook (via `uv`)

Prerequisites

- NVIDIA GPU (e.g., RTX 2050 / T4 / RTX 3060+) with CUDA 12.4+
- FFmpeg installed on system path
- `uv` package manager installed

Step-by-Step Setup

```
# 1. Clone or enter project directory
cd sentinel-ai

# 2. Add dependencies
uv add ultralytics opencv-python requests huggingface-hub pillow supabase
pyyaml pandas python-dotenv lapx

# 3. Install CUDA-accelerated PyTorch
uv pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu124 --force-reinstall

# 4. Run Pre-flight Check (downloads UVH-26 model weights & verifies
Supabase)
uv run setup.py
```

Running the Live Pipeline

```
# Run real-time detection on cam01 with desktop GUI window for 60 seconds
uv run main.py --cam cam01 --duration 60 --display

# Run on Paldi Circle (cam04) for 120 seconds in headless server mode
uv run main.py --cam cam04 --duration 120
```